

HAOYU(ALVIS) YAN

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EDUCATION

Cornell University

August 2025 - Expected May 2026

- Master of Engineering in Computer Science with a concentration in Machine Learning. GPA: 4.1/4.0

New York University

September 2021 - May 2025

- Bachelor of Arts in Computer Science and Economics. GPA: 3.9/4.0

PROFESSIONAL EXPERIENCE

Xenon Health

December 2025 - Present

Software Engineer Intern

Hybrid, Jersey City, NJ

- Built a financial record extraction system with a team of engineers using a **React** front-end and **Quart** back-end to extract and process structured information from financial documents
- Conducted benchmarking experiments and optimized a production **machine learning pipeline** for payment information extraction from PDFs, implementing a multi-stage filtering-extraction pipeline with automated accuracy tests; achieved 100% accuracy on a 481-page test set.
- Added both front-end and back-end features for billing claim creation, review, and submission. Enabled end-to-end claim processing within the platform and added a full **CI-CD pipeline** with **Azure DevOps**.

Utilyze, Inc.

December 2025 - Present

Software Engineer/Machine Learning Engineer Intern

Remote, United States

- Led the predictive maps team to build an interactive visualization platform using **MapLibre**, enabling real-time monitoring of 2000+ critical infrastructure locations and outages across service regions.
- Designed and implemented an end-to-end **data pipeline** for a predictive outage map MVP, including real-time data ingestion, processing, and dynamic visualization.
- Led sprint planning and execution for the outage maps team using **Excel**, identifying technical bottlenecks and coordinating delivery of system features.

Bimini Vision/EyeSpace LLC

April 2024 - February 2025

Software Engineer/Machine Learning Engineer Intern

Remote, United States

- Designed, developed and evaluated deep learning models with **PyTorch** and **Transfer Learning** to predict vault sizes (the space or distance between a contact lens or intraocular lens and the cornea) with OCT images (a non-invasive imaging scan of the eye) from ~200 patients. Achieved around 90% prediction accuracy within 400 microns for ResNet-based models.
- Conducted model experimentation and performance tuning across architectures and preprocessing/augmentation strategies to improve prediction accuracy.
- Developed and maintained full-stack applications with **Streamlit** for model hosting and demo.

PROJECT EXPERIENCE

Carbon Credit Trading Blockchain-Based Market

August 2025 - Present

Research Project Developer

Ithaca, NY

- Participated in the development and enhancement of a carbon credit trading market project (proof of concept) led by Xinlai Liu based on **Hyperledger Fabric Blockchain** with a full-stack architecture, leading the back-end team.
- Designed and integrated a hybrid blockchain-PostgreSQL database structure where Hyperledger Fabric serves as the source of truth for carbon credit transactions, and **PostgreSQL** acts as a queryable replica for backup.
- Implemented API services for synchronized data management and data backup/restoration functionalities.
- Implemented double-auction logic on a distributed blockchain-backed system, deployed services on **Amazon AWS**.

Bond Fund Portfolio Management with Machine Learning

May 2024 - May 2025

Research Assistant

New York, NY

- Participated in a research project based on [this paper](#) led by Luofeng Zhou to apply Machine Learning in constructing mutual fund portfolios and making data-driven investment decisions.
- Constructed deep-learning models in **Pytorch** to predict abnormal returns of bond funds using real-world time series financial data (1995 ~ 2023), built the data pre-processing pipeline using **Numpy and Pandas**.
- Evaluated model performance using financial metrics (sensitivity analysis, Sharpe ratio), visualized results using **Matplotlib**, and performed **data analysis** to identify investment opportunities and conduct risk assessments.

SKILLS

Programming Languages: Python, C/C++, JavaScript, Go, R

ML/Data: PyTorch, TensorFlow, Numpy, Pandas, Excel

Systems/Infrastructure: CUDA, SQL, PostgreSQL, Docker, Kubernetes

Full Stack: React, Quart, Next.js, FastAPI, Streamlit, LangChain, HTML

Relevant Courses: Advanced Machine Learning Systems, Cloud Computing and ML Hosting, Deep Learning